

ResilioSpace: Cloud-Native Self-Adaptive and Self-Healing 2D-to-3D Architectural Visualization Platform

CNAD Project Review-2

23th September, Wednesday



Under the Guidance of

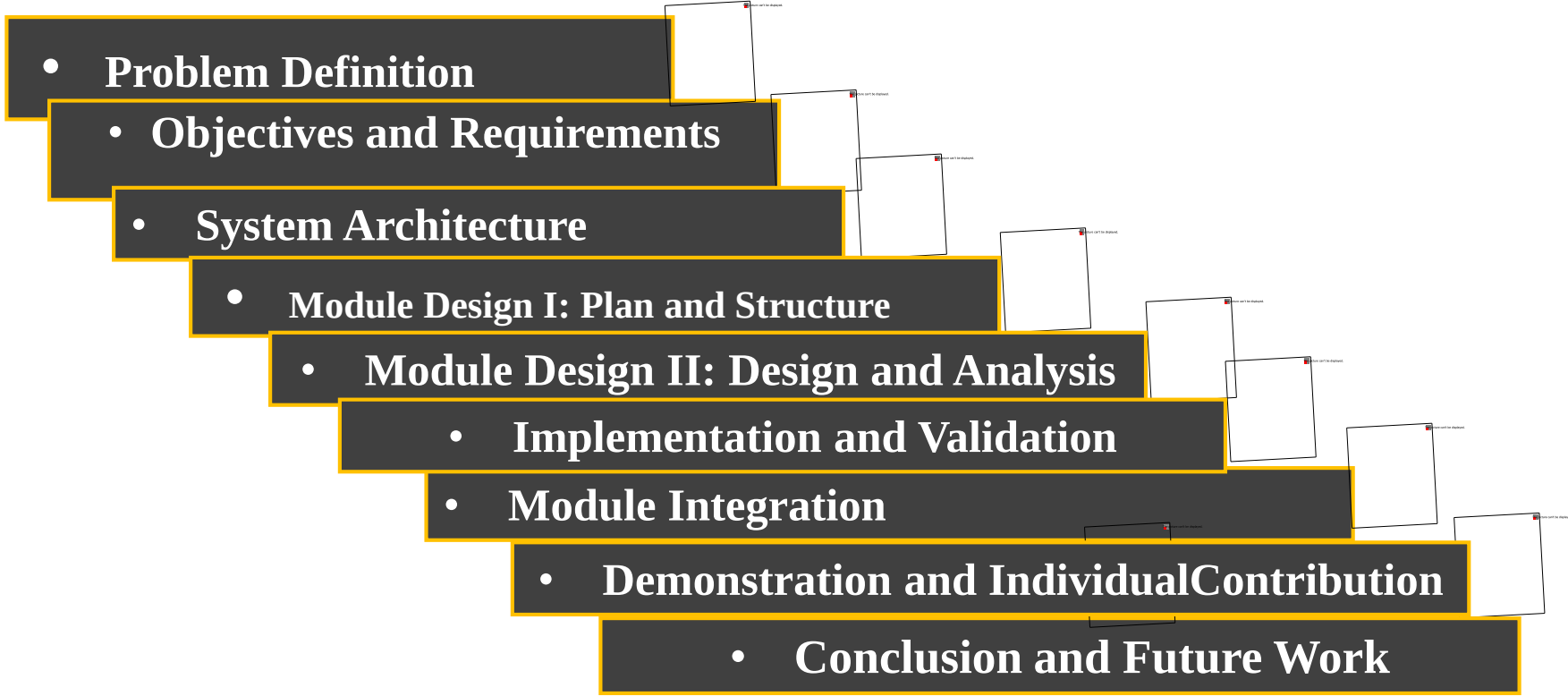
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Outline of Project Work

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- **Problem Definition**
 - **Objectives and Requirements**
 - **System Architecture**
 - **Module Design I: Plan and Structure**
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 - **Module Integration**
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Problem Definition

Manual reconstruction

- Users often redraw walls, rooms and openings before they can visualize or edit a floor plan.

Disconnected tools

- Separate applications handle reconstruction, 3D visualization, analysis and design comparison.

Failure interruption

- Renderer or processing failures can interrupt the workflow and risk loss of unsaved user state.

Defined problem

A single system should convert a floor-plan image into an editable model, synchronize 2D and 3D views, support design analysis and continue providing useful functions when individual components fail.

Proposed Solution

ResilioSpace is a web-based floor-plan reconstruction and adaptive design platform.

The system allows users to:

- Upload a floor-plan image.
- Detect walls, rooms and openings.
- View the reconstructed plan in 2D and 3D.
- Modify wall height, room types and materials.
- Save and compare different design versions.
- Perform Traditional Vastu Rule Analysis.
- Monitor application health through a Control Center.
- Adapt to failures using a MAPE-K control loop.
- Simulate controlled failures through the Resilience Lab.
- Monitor real application metrics using Prometheus.

Main Objective

To develop a usable floor-plan platform that combines reconstruction, visualization, design analysis, system monitoring and automatic recovery.

System Requirements

FUNCTIONAL REQUIREMENTS

- Upload, blank-space and starter-plan inputs
- Editable walls, rooms, doors and windows
- Synchronized 2D and 3D visualization
- Room assignments, props and materials
- Vastu analysis, versions and plan export
- Control Center, metrics and resilience simulations

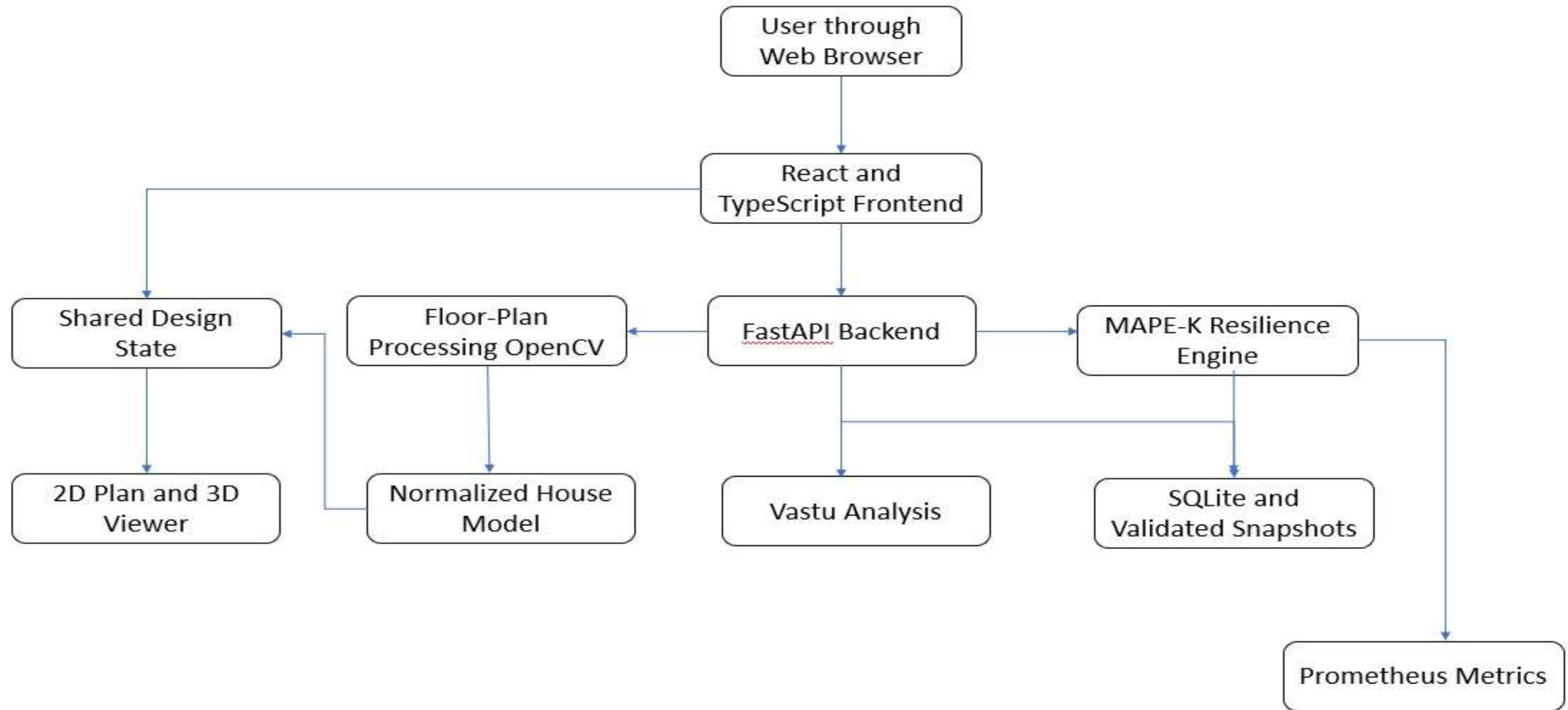
NON-FUNCTIONAL REQUIREMENTS

- Responsive browser interface
- Validated snapshots and bounded retries
- Low coupling through a normalized model
- Observable backend and renderer health
- Docker-based portability and no hardcoded secrets
- Maintainable frontend and backend tests

Technologies Used

Layer	Technology
Frontend	React, TypeScript, Vite
Production frontend server	nginx
Backend	Python, FastAPI
Image processing	OpenCV
2D representation	React-based reconstruction workspace
3D visualization	Browser-based 3D renderer
Data storage	SQLite and validated snapshots
Self-adaptation	MAPE-K feedback loop
Monitoring	Prometheus and prometheus_client
Containerization	Docker and Docker Compose
Testing	Pytest and frontend test framework
Version control	Git and GitHub

System Architecture



Module Design I: Plan and Structure

1. Input Options

- **Upload Floor Plan:** User can upload an existing floor-plan image.
- **OpenCV Processing:** The uploaded image is processed to identify the basic structural elements.
- **Blank Plan:** User can start a new floor plan by defining required dimensions.
- **Sample Plan:** A predefined sample layout is provided for quick testing and demonstration.

2. Structure Editor

➤ Walls

- Create new walls.
- Move and reposition existing walls.
- Modify the basic structure of the floor plan.

➤ Rooms

- Add new rooms to the plan.
- Resize rooms according to requirements.
- Assign room types or names such as bedroom, kitchen, hall, etc.

➤ Openings

- Add doors and windows to walls.
- Attach openings to specific walls.
- Move openings along the wall while maintaining their position within the structure.

➤ Export

- Convert the completed design into a clean 2D structural image.
- Download the final floor-plan PNG for further use.

Module Design II: Design and Analysis

1. Design Studio

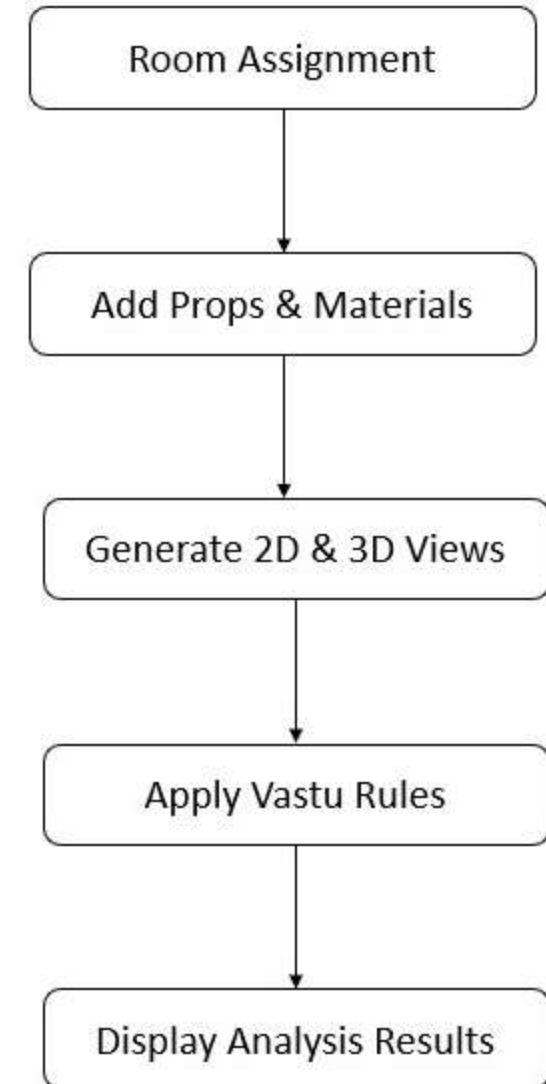
- Assign **names and types** to different rooms.
- Add **furniture and interior elements**.
- Add **materials and wall-mounted objects**.
- Move and rotate objects according to the room layout.
- Provides controlled editing without disturbing the overall structure.

2. 2D and 3D Views

- Provides both **2D floor-plan and 3D visualization**.
- Both views are generated from the same underlying design data.
- Changes made in one view are reflected in the other.
- Represents **wall height and room structure**.
- Displays doors, windows, and other openings.
- Keeps furniture and object movements synchronized.

3. Traditional Vastu Analysis

- Analyzes the floor plan based on configurable Vastu rules.
- Considers **room type and room location**.
- Uses the **centre of the room/plan** for analysis.
- Considers **North direction and orientation**.
- Generates applicable Vastu-related results based on the configured rules.



MAPE-K and Resilience Lab

Stage	Responsibility
Monitor	Collects renderer, processing and application observations
Analyze	Determines whether the system has a fault or unhealthy condition
Plan	Selects a suitable adaptation strategy
Execute	Applies the strategy to the working application
Knowledge	Stores policies, previous events and adaptation outcomes

Resilience Lab

The Resilience Lab injects safe simulated conditions into the existing MAPE-K loop.

The simulations are:

- Local
- Bounded
- Reversible
- Clearly marked as simulated
- Designed to preserve user data

The project does not create a separate fake adaptation engine. Simulations use the same Monitor, Analyze, Plan, Execute and Knowledge components as normal application behavior.

Module Integration

1. Plan Input

- Start with an **uploaded, blank, or sample** floor plan.
- Provides the initial design information.

2. Structure

- Organizes the plan into **walls, rooms, and openings**.
- Structural changes are maintained in a common model.

3. Shared Model

- Maintains a **single normalized design state**.
- Keeps all modules synchronized and consistent.

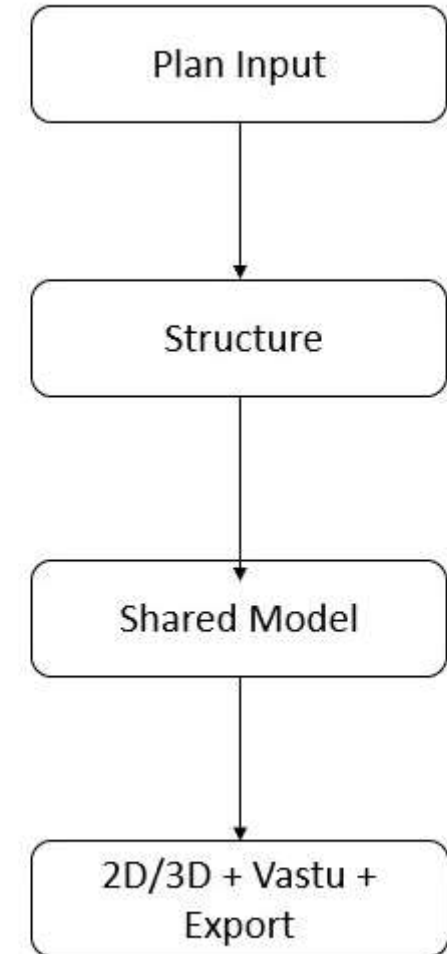
4. Consumers

- **2D View** – floor-plan visualization.
- **3D View** – 3D representation.
- **Vastu** – rule-based analysis.
- **Export** – final design output.

Example: Door Synchronization

Save → Render 2D → Render 3D → Move

- Door details are stored in the shared model.
- Moving the door updates its position in **both 2D and 3D views**.



Implementation and Validation

1. Testing Summary

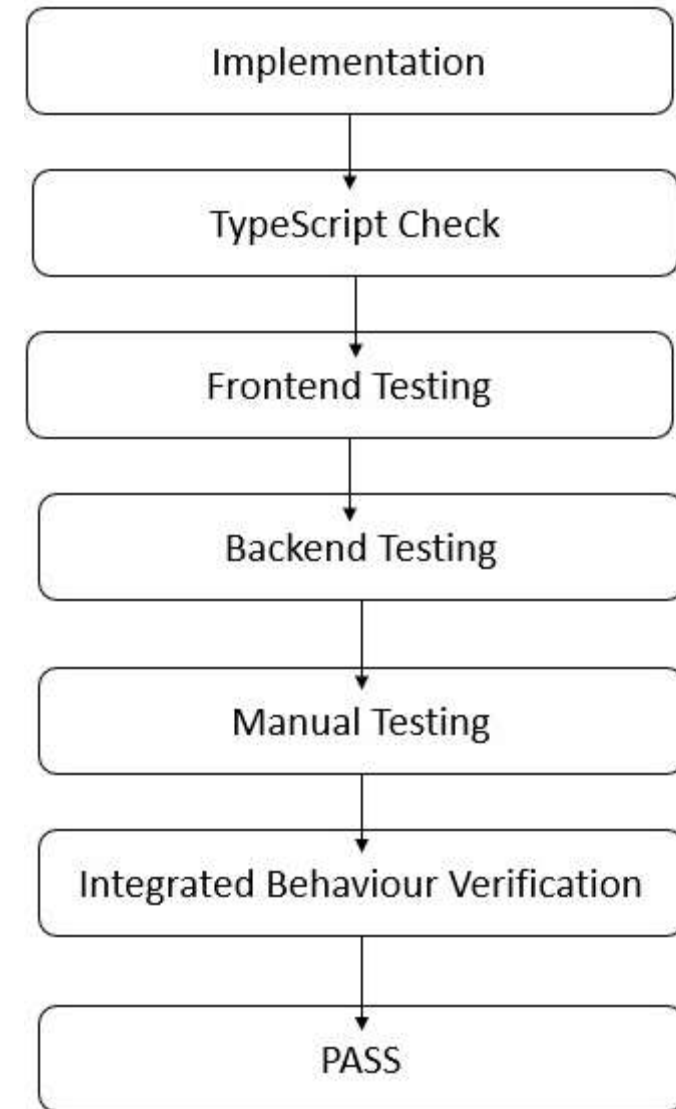
- **30 Frontend enhancement tests** – Passed
- **10 Props & Vastu tests** – Passed
- **8 Opening geometry tests** – Passed
- **1 Bathroom Vastu backend test** – Passed
- **Total: 49 focused tests – All Passed**

2. Verified Behaviours

- Blank-plan dimensions correctly generate the **physical geometry**.
- Doors and windows remain **associated with their walls**.
- Wall decorations remain **inside the wall constraints**.
- Vastu analysis filters irrelevant rules and supports **Bathroom-specific rules**.

3. Engineering Validation

- **TypeScript:** Type checking passed.
- **Frontend:** Vitest test suites passed.
- **Backend:** Pytest test cases passed.
- **Manual Testing:** 2D/3D views, Vastu analysis, simulation, and recovery verified.



Demonstration and Individual Contribution

- **Start** – Show the three plan input options.
- **Create** – Enter dimensions for a blank plan.
- **Structure** – Add rooms, doors, and windows.
- **Design** – Assign room types and add props.
- **Visualize** – Compare synchronized 2D and 3D views.
- **Analyze** – Perform Vastu analysis using North orientation.
- **Recover** – Demonstrate failure simulation and state recovery.

Individual Contributions

- **Masetti Manasvini:** Floor-plan reconstruction, walls, rooms, doors/windows.
- **Doma Akshaya:** 2D/3D design, room classification, props and Vastu analysis.
- **Nallapu Sreja:** MAPE-K, monitoring, resilience and recovery.
- **All members:** Integration, testing, debugging, and final validation.

Conclusion and Future Work

Current Limitations

- Uploaded floor plans may need **manual corrections**.
- PNG export does not preserve **editable design data**.
- Vastu analysis is a **configurable cultural reference**

Future Work

- Add **project import/export** for editable continuation.
- Improve **computer-vision-based reconstruction**.
- Add **cloud storage and collaboration**.
- Expand the **object/material library**.

Conclusion

ResilioSpace integrates floor-plan creation, synchronized 2D/3D design, Vastu analysis, monitoring, and resilient operation using a unified house model.